Weather → Revenue · Roofing Portfolio

Altis Groep challenge · companies: Peter Ummels (Brunssum) + Gilde GB (Valkenswaard) · 2023–2026

Open-Meteo daily weather (100% coverage) vs invoice-dated ledger revenue (€80.2M combined)

Key findings

✓ HEADLINE Each extra WORKABLE day/week ≈ +11–14% revenue
(workable = no rain, frost, heat ≥28°C or wind ≥40km/h).
Robust across regression + pooled 2-company panel (p≈0.03).

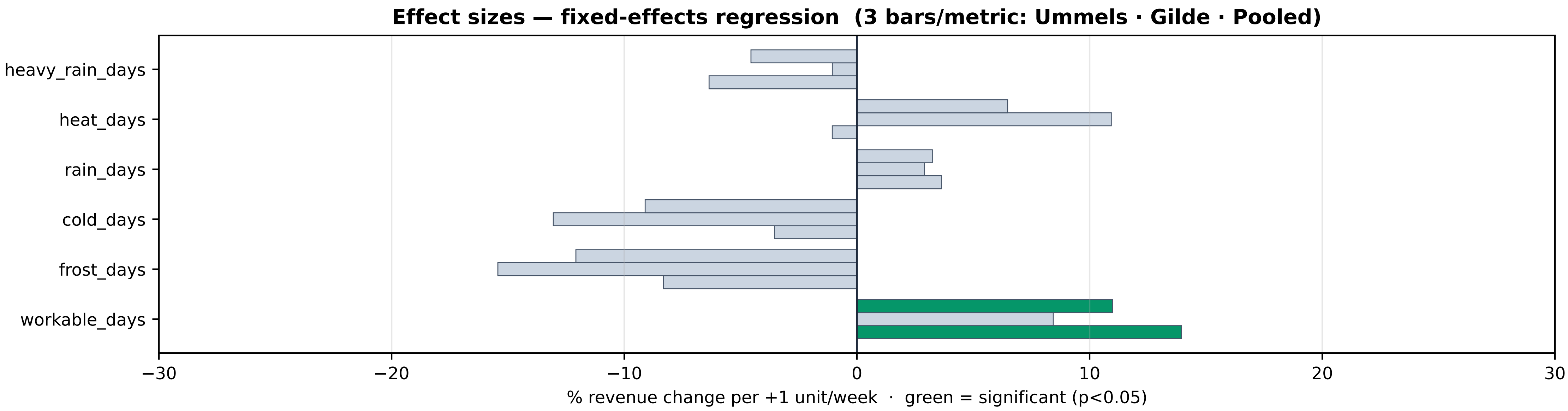
~ SECONDARY Frost/cold drag revenue — strong for Gilde, mostly
seasonal for Ummels. A real winter dip, hard to call causal.

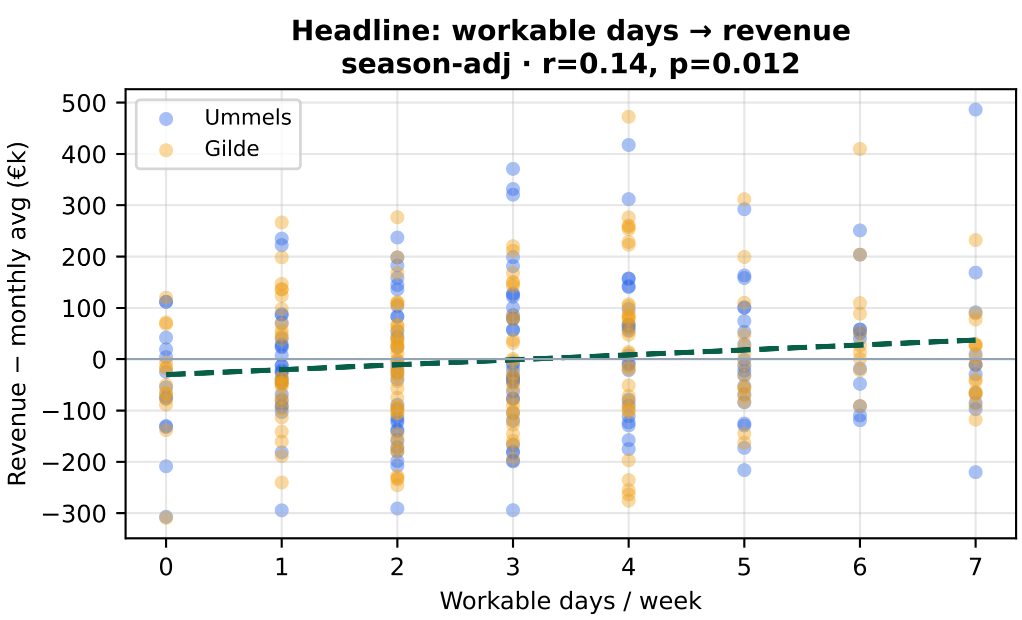
✗ NO SIGNAL Rain, heat, wind ALONE: nothing once season removed.

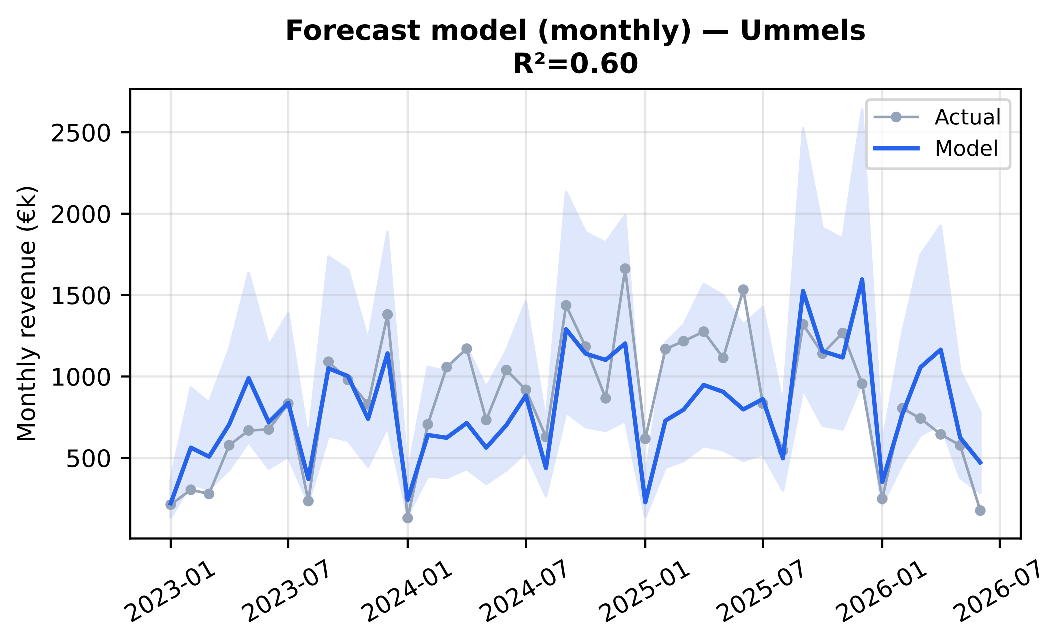
! LIMIT All data is INVOICE-dated, not work-dated; no hours.
Invoice timing blurs day-level weather effects.

Data assets

Source	Size	Date type	Weather-ready
Ummels (Brunssu	€35.8M · 9.8k txns	invoice	✓ weather
Gilde (Valkenswaa	€44.4M	invoice	✓ weather
Altis dataset 1	monthly turnover	month	no location
Altis dataset 2	txns + Company E	invoice	no location
Open-Meteo	8 vars, daily	daily	✓ 100% cover







What we can do with this

- Quantify weather sensitivity per company (done) — workable-days elasticity is the headline KPI.
- Weather-rule playbook (output_playbook/): backtested rules; only Gilde frost-week survives season-adjustment.
- Revenue forecast tool (output_forecast/): feed a weather forecast → expected monthly revenue + 80% band (R²≈0.60).
- Seasonal planning: model & anticipate the winter frost/cold revenue dip.
- Benchmark companies: Gilde is more frost-sensitive than Ummels.

Biggest unlock → obtain actual WORK/JOB dates or labour hours; then daily causation becomes provable.